

14897 Topics in Mobile Computing

Mini Project

Spring 2026

The mini project will provide students with hands-on experience in [mobile computing and networking systems](#) covered in the course. Each student has to carry out a mini project individually. There are two options for you to choose from:

- (a) Performance comparison of two or more algorithms, protocols, software or systems (which are related to mobile computing and networking) using *mathematical analysis*, *computer simulation* or *real measurement*: A new design of algorithm, protocol, software or system is also welcome but a performance comparison with existing ones must be included.
- (b) Writing a survey paper on an emerging topic in *mobile computing and networking systems*: Qualitative and/or quantitative comparison of various schemes/techniques in your topic must be addressed and discussed by yourself.

All project topics must be approved by the instructor.

1. Schedule

• Project proposal

- Due on [March 18](#) (Wed)
- You may choose a project topic from a list of sample topics or suggest your own topic related to the course materials. Each student must submit a project proposal.
- Your proposal is required to **clearly state** the followings in sequence: (a) project title, (b) the problem you are going to address, (c) precise goals of the project, (d) deliverables of the project, (e) the issues you are going to tackle, (f) the project schedule (Use a Gantt chart and specify the milestones you plan to achieve on the timeline in order to accomplish the project goals and deliverables.), and (g) a list of relevant literature cited.
- Note that your project topic must be approved by the instructor.

• Progress report

- Due on [April 08](#) (Wed)
- You are required to submit a progress report summarizing work completed to date. The report must clearly describe the specific problem you have been working on. It must also include any modifications to the project plan (such as an updated Gantt chart).

- The progress report and the final report must be in the *research paper format*. That is, they must conform to the IEEE conference paper templates (which are available at http://www.ieee.org/conferences_events/conferences/publishing/templates.html) or equivalent.

- **Presentation**

- In class on **May 27** (Wed)
- All the students must give an oral presentation on their project. A working prototype will be demoed if your project is of the sort that can be demonstrated. Due to the shortage of time, some presentations may be held out of class time if required.

- **Final report**

- Due on **June 3** (Wed)
- You are requested to submit the final report. The contents of your report are described in the “Suggested topics and notes.”
- Note that the final report must be in the *research paper format* as mentioned in the progress report. In addition, if you write a survey paper, follow the typical survey paper style in terms of organization and presentation. (You may refer to some existing survey papers published in high-quality journals as well.)

2. Suggested topics and notes

- The following lists are some examples that are intended to get you thinking about possible project ideas. You can suggest a topic related to the class materials. You also search interesting topics from the Internet. You will produce a much better idea and enjoy it.
- Performance comparison of two or more algorithms, protocols, software or systems (which are related to mobile computing and networking) using *mathematical analysis*, *computer simulation* or *real measurement*: **A new design of algorithm, protocol or system is also welcome but a performance comparison with existing ones must be included.** The contents of your report may include title, authors, abstract, keywords, introduction, related works, main sections (studied ones of you), performance evaluation, conclusions, and references. ***More than 10 references* must be cited in your report, and every reference must be cited at least once in the report.** The followings are some example topics:
 - Your own design on mobile computing (algorithm, protocol, software, systems, etc.)
 - Your own design which is related to mobile computing areas
 - Relative or derivative topics on the readings of the course

- Energy efficiency in TCP/routing/MAC protocols for mobile computing & networks
 - Network performance in TCP/routing/MAC protocols for mobile/wireless networks
 - Impact of reinforcement learning-based algorithms in mobile computing & networks
 - Cross-layer optimization and performance tuning in emerging mobile systems
 - Operating system and middleware issues to support mobile IoT and edges
 - Algorithms and complexities in mobility tracking for vehicular ad hoc networks
 - Performance of routing/MAC protocols in unmanned aerial vehicle (UAV) networks
 - Performance of resource management algorithms in mobile edge computing
 - Security and privacy studies for mobile computing and networking
 - Performance study of mobile computing issues in generative AI or physical AI
 - Performance of emerging mobile applications
- Writing a survey paper on an emerging topic in *mobile computing and networking systems*:
 In this case, you must investigate related works done so far in the world, and give your own analysis and comparison with technical evaluation. The contents of your report may include title, authors, abstract, keywords, introduction, main sections (surveyed ones of you), qualitative and/or quantitative comparison, challenging issues, conclusions, and references. More than 20 references must be cited in your report, and every reference must be cited at least once in the report. It is suggested that the length of the survey paper is 8~15 pages or more in the technical paper format. The followings are some example topics:
- Smartphone sensing and its applications in hyper-connected society
 - Energy conservation techniques in emerging mobile computing & networks
 - Operating systems/middleware for mobile computing
 - Reinforcement learning-based algorithms for mobile computing systems
 - Security and privacy in body area sensor computing & networks
 - Various mobility models for hybrid air-ground vehicular networks
 - Opportunistic medium access in mobile IoT networks
 - Heterogeneous mobile networks with smart antennas
 - Scheduling and MAC protocols for unmanned aerial vehicle (UAV) networks
 - Energy-efficient resource management in mobile edge computing systems
 - Communication protocols for time-critical applications in UAV-enabled IoT
 - Task offloading mechanisms in emerging mobile IoT and edges
 - Emerging topics of mobile computing systems in generative AI or physical AI
 - Relative or derivative topics on the readings of the course

3. Grading criteria

- Clarity: Does it clearly state what to develop/solve and what have been done?
- Originality: Is the proposed idea/solution original?
- Approach: How much realistic are the development process and experiment environment?
- Presentation/Demo: Are the presentation and demo persuasive?
- Report: Are the reports (proposal, progress report, and final report) nicely organized and well presented?

4. Important remarks

- Submit your work at the course webpage *by the specified deadline* (or *by 23:59 on the due date* if not specified).
- When your work is not turned in on time, a late penalty will be given.